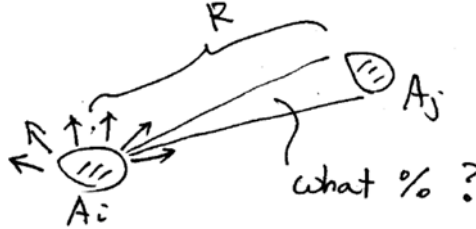


29. View Factor (Configuration Factor)

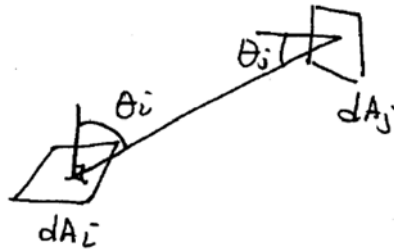
[Definition]

$F_{ij} \equiv$ "Fraction of radiation leaving surface i that is **intercepted** by surface j "



Intuitively,

$$F_{ij} = f(A_i, A_j, R_i, \theta_i, \theta_j) \quad \& \quad 0 \leq F_{ij} \leq 1$$



$$dq_{i \rightarrow j} = I_i \cos \theta_i dA_i d\omega_{i \rightarrow j} = I_i \frac{\cos \theta_i \cos \theta_j}{R^2} dA_i dA_j$$

In the case of diffuse surfaces, $J_i = \int_{\text{Hemisphere}} I_i \cos \theta d\omega = \int_0^{2\pi} \int_0^{\pi/2} \cos \theta \sin \theta d\theta d\phi = \pi I_i$

$$dq_{i \rightarrow j} = J_i \frac{\cos \theta_i \cos \theta_j}{\pi R^2} dA_i dA_j$$

Integration yields $q_{i \rightarrow j} = J_i \int_{A_i} \int_{A_j} \frac{\cos \theta_i \cos \theta_j}{\pi R^2} dA_i dA_j$

Also, $q_{i \rightarrow \text{Hemisphere}} = A_i J_i$

$$\Rightarrow F_{ij} = \frac{q_{i \rightarrow j}}{q_{i \rightarrow \text{Hemisphere}}} = \frac{1}{A_i} \int_{A_i} \int_{A_j} \frac{\cos \theta_i \cos \theta_j}{\pi R^2} dA_i dA_j$$

(Note) $F_{ji} = \frac{1}{A_j} \int_{A_j} \int_{A_i} \frac{\cos \theta_i \cos \theta_j}{\pi R^2} dA_i dA_j$ and thus

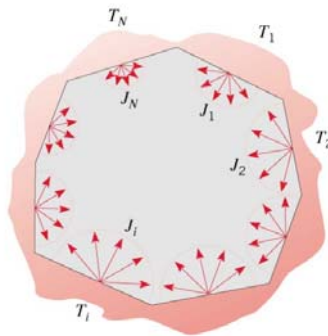
● Reciprocity of view factor relation $A_i F_{ij} = A_j F_{ji}$

Ex) $A_1 F_{12} = A_2 F_{21} \rightarrow$ If $A_1 = A_2, F_{12} = F_{21}$

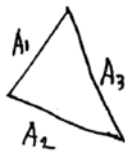
If $A_1 \gg A_2, F_{12} = \frac{A_2}{A_1} F_{21} \approx 0 \quad F_{21} = \frac{A_1}{A_2} F_{12} \approx 1$

- Summation rule: For enclosure of N surfaces $\sum_{j=1}^N F_{ij} = F_{i1} + F_{i2} + \dots + F_{iN} = 1$

Note) $F_{ii} \neq 0$ in general (e.g., concave surface)



ex)



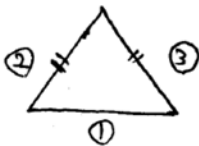
$F_{11} = 0$ (flat or convex)

If $F_{12} = 0.4, F_{13} = 0.6 \therefore F_{11} + F_{12} + F_{13} = 1$

- Symmetry relation: For enclosure of N surfaces

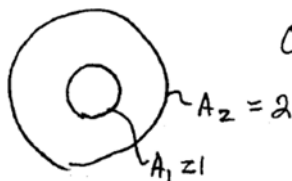
Using symmetry, F_{ij} can be obtained intuitively $\sum_{j=1}^N F_{ij} = F_{i1} + F_{i2} + \dots + F_{iN} = 1$

ex)

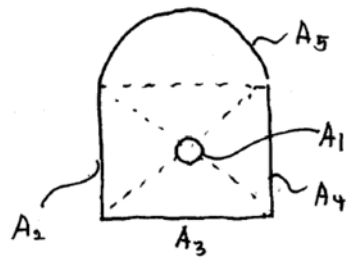


$F_{12} = F_{13}$ by symmetry

Then, from summation rule $F_{12} = F_{13} = 0.5$



$F_{11} = 0 \quad F_{12} = 1 \quad F_{21} = \frac{A_1}{A_2} F_{12} = \frac{A_1}{A_2}$
 $\therefore F_{22} = 1 - \frac{A_1}{A_2} = 0.5$

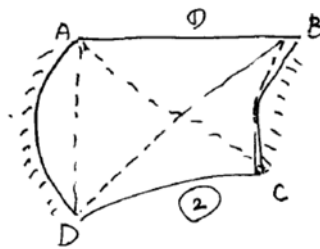


Q) F_{51} ?

By symmetry, $F_{15} = \frac{1}{4}$

$$\therefore F_{51} = \frac{A_1}{A_5} F_{15}$$

- Hottel's cross-string method for 2-D view factors (1967)



$$F_{12} = \frac{\sum (\text{crossed strings}) - \sum (\text{Uncrossed strings})}{2L_1} = \frac{\overline{AC} + \overline{BD} - \overline{AD} - \overline{BC}}{2L_1}$$

(Note) AD, BC: length of strings stretched tightly between the respective points